
DTU

Mathematical Modelling 02526

Salami Feedback

Overall Impression

Most groups understood the core task and produced working implementations.

However, the range in report quality was very wide…

~27/35

Had clear structure (Intro/Method/Results/ Discussion/Conclusion)

4/35

Included references to sources

~30/35

Had a separate Discussion section

Report Structure

Does the report follow the classical scientific format?

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Expected: Introduction → Data & Methods → Experiments & Results → Discussion → Conclusion

Good examples:

- Clear section headings
- table of contents, logical flow
- One report with a Related Work section

Problems observed:

- Several reports: No Discussion section, jumps from results to conclusion.
- 1 report: No section headings at all, everything merges into flowing text.
- 2-3 report: Sections exist but are extremely brief, methods and data on a single page.

Introduction

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(2) describe WHAT will be done, and
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General remarks:

- **Lack of purpose**
- **Why use both a very simple and a more complex method?**
- **Several groups copy the exercise text's background verbatim rather than writing their own**

Methods and Model description

Is the model clearly stated? Could you reproduce the results?

The methods section should: describe the model, list assumptions, define all symbols, and provide enough detail for reproduction.

General remarks:

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- **Why use both a very simple and a more complex method?**
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Methods and Model description

Single-band threshold model. For a fixed band, each pixel is represented by an intensity x. We consider two classes C_f (fat) and C_m (meat) and use the threshold rule

$$\tau(x) = \begin{cases} C_f, & x \geq t, \\ C_m, & x < t. \end{cases}$$

For each class we assume a normal distribution with a common variance, $x \mid C_k \sim \mathcal{N}(\mu_k, \sigma^2)$, $k \in \{f, m\}$ which gives the threshold is

$$t = \frac{\hat{\mu}_f + \hat{\mu}_m}{2}.$$

This procedure is repeated for each of the 19 bands and the band with the lowest error on the annotated pixels is selected.

Linear Discriminant Analysis (LDA). For the multispectral model, each pixel is represented by a vector $x \in R^n$ with $n = 19$. We assume two multivariate normal distributions with shared covariance,

$$x \mid C_k \sim \mathcal{N}(\mu_k, \Sigma), \qquad k \in \{f, m\}.$$

The class mean vectors μ_f and μ_m are estimated by sample averages of the annotated pixels in each class, and the inverse pooled covariance matrix is used for classification:

$$S_k(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_k - \frac{1}{2} \hat{\mu}_k^T \hat{\Sigma}^{-1} \hat{\mu}_k + \ln p_k, \qquad k \in \{f, m\},$$

where p_f and p_m is the prior probability.

We can define a decision function by forming the difference $g(x) = S_f(x) - S_m(x)$. Using the expression above, this gives

$$g(x) = x^T \hat{\Sigma}^{-1} (\hat{\mu}_f - \hat{\mu}_m) - \frac{1}{2} \left(\hat{\mu}_f^T \hat{\Sigma}^{-1} \hat{\mu}_f - \hat{\mu}_m^T \hat{\Sigma}^{-1} \hat{\mu}_m \right) + \ln \frac{p_f}{p_m}.$$

- Separates single-band and LDA models clearly
- States assumptions
- Defines all variables

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- States assumptions
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- Why is it a good assumption?
- Nearly all state the assumptions of normality
- Very few discuss whether it is met and what the implications of that is
  - If it is not a problem, then argue why
  - If it is a problem, then come with suggestions for what to try instead

Methods and Model description

The model we're using is a Multi-variant Linear Discriminative analysis also known as LDA. It works by using every pixel in a multispectral picture. With our salami it classifies every pixel as either meat or fat. The model works by taking a pixel, identify it, and then distribute it into its class by the highest discriminate value.

First the model declares each pixel as an independent multi-variant variable, which corresponds to the 19 spectral bands. It then also assume that each observation in each class follows a multivariable normal distribution, and that they have a common pooled covariance matrix between the two classes. The classification then happens by the discriminate function:

$$S_i(x) = x^T \hat{\Sigma}^{-1} \hat{\mu}_i - \frac{1}{2} \hat{\mu}_i^T \hat{\Sigma}^{-1} \hat{\mu}_i + \ln p_i$$

We know that the LDA works because of the way the salami absorbs light. The meat absorbs blue and yellow light which is why it has the red colour and the white fat appears white because it absorbs little to no light across all wave lengths.

To get a high discriminative value instead of a simple thresholder, we use 19 bands to get a correlation between the wave lengths. We then used a pooled covariance matrix to estimate the model.

The model works because of it's physical and spectral properties of the materials.

Even though the LDA is quite effective it also has some limitations. The LDA forces the data to be linear because it assumes that both classes have the same covariance matrix. Because of this it sees every outlier from the normal distribution to be noise and therefore tries to remove it.

This is a complication because we use this model not only on the day one salami but all the way to the day 28 salami. The meat of the salami will change its colour over time which therefore also kills the precision over time.

- Data, methods, and formulas **crammed together**
- **No single-band** threshold model shown; Jumps straight to LDA
- **One equations** is presented **without full context**. Hard to reproduce.
- **Not explaining** variables
- Paragraphs with one sentence
- **Spelling/grammar** mistakes, strange phrases like the model works

Data

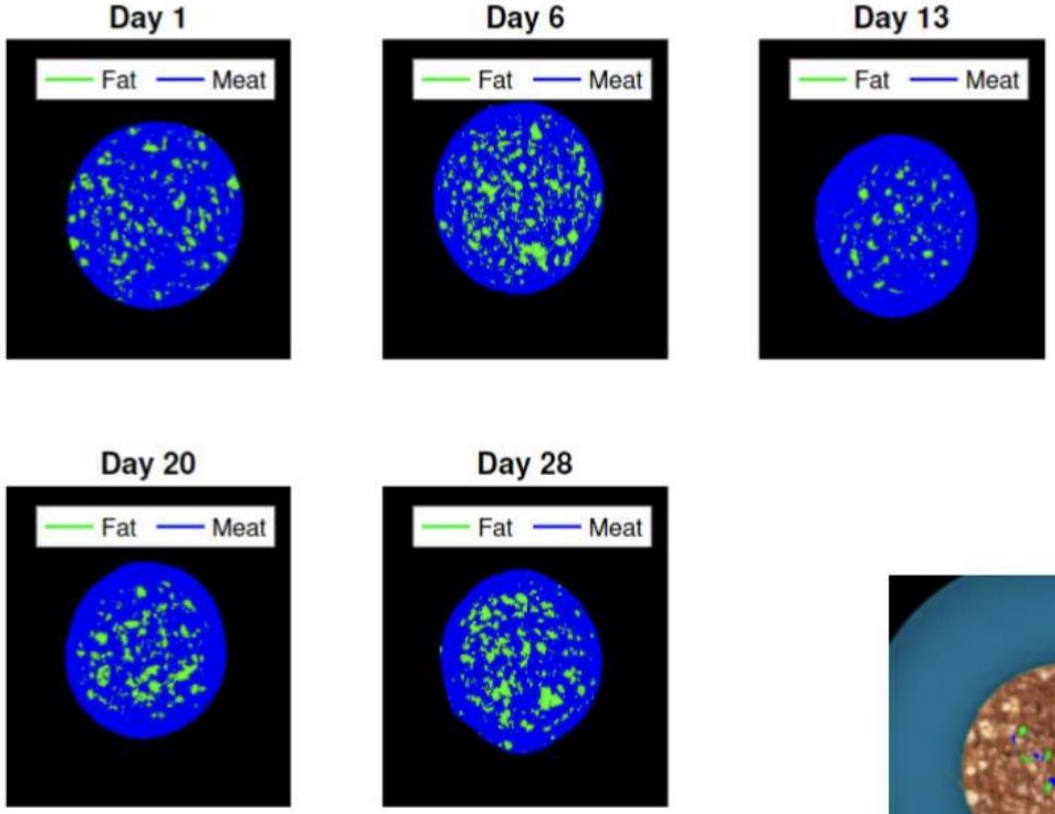
Where do the data come from?

We are working with spectral images of the salami over the 28 days of drying.

The experiment is using the VideometerLab to take a picture of a sliced salami in 19 different spectral bands on day 1, 6, 13, 20 and 28. This is done to see how the salami changes over the days and is used to train an algorithm to spot meat and fat.

The provided data for this project are 5 multi spectral images of the cross-section of the same salami, taken at days 1, 6, 13, 20 and 28 during the aging process, each with 19 images of the sausage at different wavelengths. In addition to that, food sci-

Figures and Visualizations



- Think about the reader
- The reader has (most likely) not seen the data before
- What is the context of your results?

- Always cite a figure in the main text!

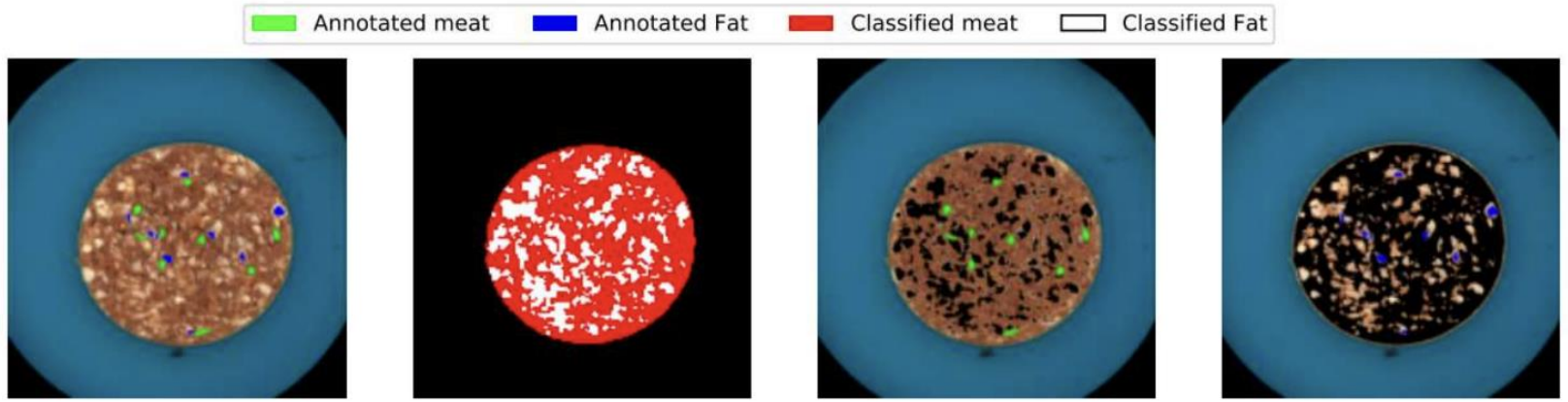


Figure 2: From left to right we see the day 1 image, the classification, the image with the removed fat(black), and the image with the removed meat(black).

Figures and Visualizations

- Reports with no images!!
- Reports without showing the data
- Missing caption or very short captions
- Missing color labels, legends (which one is meat/fat?)
- Some figures are too small to read when printed

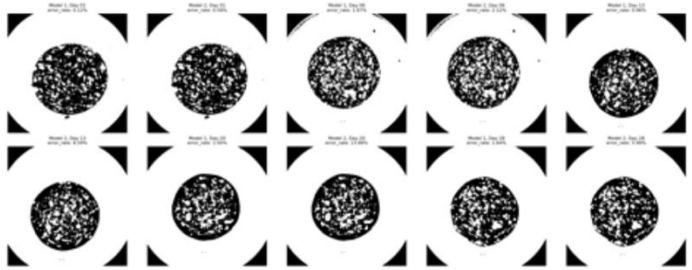


Figure 3: Visual Comparison of predictions made by both models

A

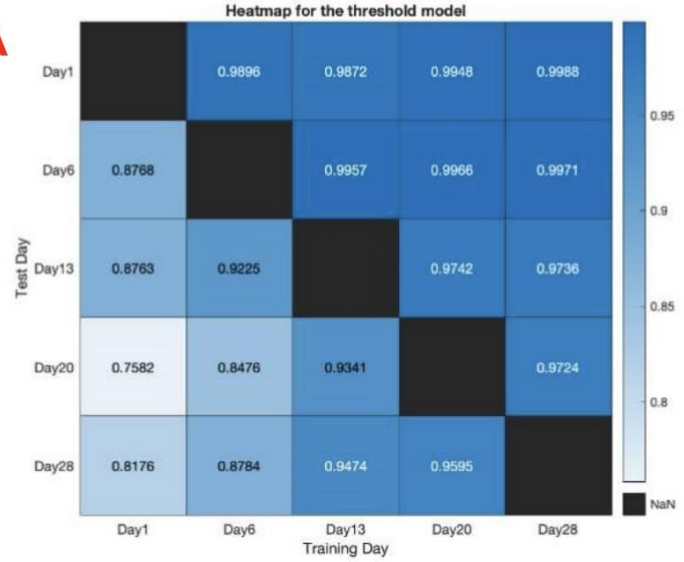
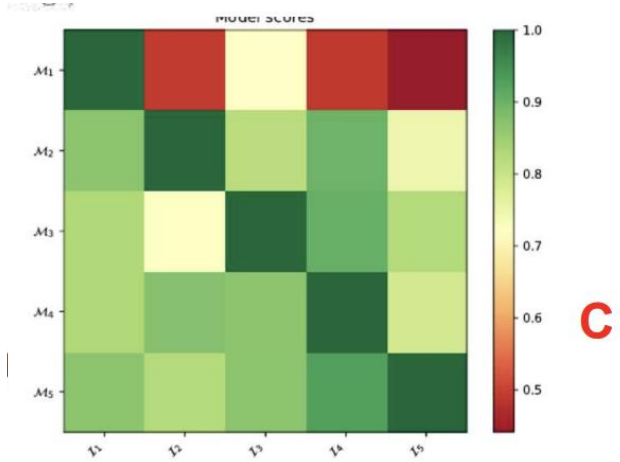


Figure 1: Success rates of the threshold classification of fat and meat for different training and test images.



C

Figure 3: This figure shows the model performance scores as described in **Experiment and results**. Model $\mathcal{M}_1$ is trained on the the annotations from day 1, model $\mathcal{M}_2$ o the the annotations from day 6, and so on. A perfect score means that the mode produced a segmentation identical to the ground truth image.

0.5797	2.319	1.101	1.101	0.696
2.117	**0.1925**	0.1444	1.107	2.310
8.589	1.562	**0.0601**	0.3604	1.021
13.88	4.235	2.000	**0.1765**	0.1176
5.985	4.247	1.496	4.923	**1.014**

Table 2: Error matrix with no prior knowledge. The diagonal is the error rate. Rounded to four significant figures.

B

D

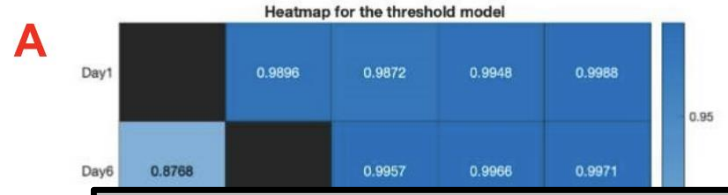
Test data \ Train data	day 01	day 06	day 13	day 20	day 28
day 01	0.005797	0.023188	0.011014	0.011014	0.006957
day 06	0.021174	0.001925	0.001444	0.011068	0.023099
day 13	0.085886	0.015616	0.000601	0.003604	0.010210
day 20	0.138824	0.042353	0.020000	0.001765	0.001176
day 28	0.059846	0.042471	0.014961	0.049228	0.010135
SSE (train data error excluded)	0.175152	0.066175	0.027336	0.051771	0.026222

Table 1: Error rates (no prior knowledge)

E

Train ↓ \Test →	Day 01	Day 06	Day 13	Day 20	Day 28	Avg. Error
Day 01	0.58%	2.17%	8.77%	14.00%	6.32%	**7.82%**
Day 06	2.43%	0.24%	1.68%	4.59%	4.54%	**3.31%**
Day 13	1.22%	0.19%	0.12%	2.71%	1.83%	**1.49%**
Day 20	1.10%	0.82%	0.36%	0.24%	5.16%	**1.86%**
Day 28	0.58%	1.78%	0.78%	0.29%	1.16%	**0.86%**

Table 3: Annotation Error Rates (%)



0.5797	2.319	1.101	1.101	0.696
2.117	**0.1925**	0.1444	1.107	2.310
8.589	1.562	**0.0601**	0.3604	1.021

- ***Always*** *cite table in text*
- *How many significant* ***digits****?*
- *What is the* ***error rate****?*
- *How* ***easy*** *is it to read the results?*
- *Make sure to give* ***proper context***
- *Should be understandable be* ***captions alone***

Figure 1: Su
and test ima

20	day 28
014	0.006957
068	0.023099
604	0.010210
765	0.001176
228	0.010135
771	0.026222

						Avg. Error
						7.82%
						3.31%
						1.49%
Day 20	1.10%	0.82%	0.36%	0.24%	5.16%	**1.86%**
Day 28	0.58%	1.78%	0.78%	0.29%	1.16%	**0.86%**

Table 3: Annotation Error Rates (%)

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Discussion and Conclusions

Are results interpreted, not just reported?

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Good discussions and conclusions (~10 reports):

- Explain **WHY** later training days give lower error (drying changes spectral properties)
- Discuss the two-class limitation (no background class).
- Note edge/casing issues. Suggest future work
- 1 report circles back to the three goals from the intro.
- 1 report discusses calibration and representative training.
- 1 report states a clear hypothesis and evaluates it.

Problems:

- No discussion section at all (at least 5 reports)!
- A couple of reports: Discussion is 3-4 lines.
- Several reports do not explain **WHY** LDA outperforms threshold.
- Common pattern: restating numbers ('error was 2.1%') without interpretation.
- No conclusion at all!

References

- Only **four** reports (!) have references
- **Pro:** You do not need to argue/explain; someone has done the job for you
- **Should not be footnotes**! Unless they are not proper references (e.g. a website).

  And for LDA you can definitely find a proper reference!
- **Your job**: to find references to support your claims, argue for your methods, etc

Key takeaways

1	**Follow the structure**	Introduction → Method → Results → Discussion → Conclusion. Always.
2	**State a clear goal**	The introduction must say what the report aims to achieve.
3	**Define everything**	Every variable in every equation. Every color in every figure.
4	**Discuss, don't just report**	WHY does LDA beat threshold? WHY does training day matter?
5	**Respect the page limit**	5 pages forces clarity. Code goes in the appendix.
6	**Include references**	Cite references, software, and any AI tools used.
7	**Write a report, not a notebook**	Prose narrative with clear sections, not reporting numbers
8	**Proofread!**	Check headers, NaN values, figure captions, and spelling.

Questions??

Questions and Discussion

Feedback on the exercise itself?

Was the exercise clear? Too long? Anything confusing?

Feedback on the peer review process?

Was the peer review useful? Too time-consuming? Suggestions?

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